The Climate Confidence Gap: Do Public Claims Match the Assessments?

Abstract

A public claim can sound more certain than the scientific assessment it cites. This paper proposes a way to check: match the claim to the assessment's actual finding, date, conditions and uncertainty. Six climate topics illustrate the method. The comparison recognizes scientific progress, including a narrower assessed climate-sensitivity range, while distinguishing it from certainty about every projection or event. The contribution is an auditable reporting method. The examples motivate a systematic study; they do not establish how frequently public communication overstates the science.

1. The Standard We Are Applying

A scientific assessment and a public statement about it should make the same claim at the same level of certainty. This paper compares the two directly: what finding is reported, what conditions apply, and what uncertainty survives in the retelling?

A scientific assessment and a public summary should convey the same result and level of certainty. Checking that match starts with reading the two claims together. The IPCC uses defined terms for confidence and likelihood; these express different aspects of the assessment and should retain their meanings when communicated to the public.

Where the public claim carries higher confidence than the source, something has been added in transmission. Where it omits a stated uncertainty, something has been removed. Both are checkable and neither requires taking a position on the underlying physics.

We examine six cases. In each we state the assessment finding, the public claim, and the size of the gap.

2. Equilibrium Climate Sensitivity

The Charney Report of 1979 gave the equilibrium warming from doubled CO as 1.5 to 4.5 degrees. The range was described as reflecting genuine uncertainty about feedbacks, particularly clouds.

AR5, thirty-four years later, gave 1.5 to 4.5 degrees. AR6 narrowed the *likely* range to 2.5 to 4.0 with a best estimate of 3.0, drawing on paleoclimate and process understanding rather than models alone, which is a real advance and should be reported as one.

What has not happened is the collapse of the range to a point. Forty-five years of work, enormous computational advance, and vastly better obser-

vation have produced a likely range still spanning a factor of 1.6. That is a statement about how hard the problem is, and it is a legitimate scientific finding rather than an embarrassment.

Confidence that humans cause warming is distinct from precision about its magnitude. A remaining sensitivity interval is compatible with well established conclusions. A criticism of communication must identify the exact proposition asserted and compare it with the relevant assessment.

3. Scenarios Treated as Forecasts

RCP8.5 was constructed as a high-emissions pathway, requiring coal use to expand severalfold through the century. It was widely labelled business-as-usual in the literature and almost universally in press coverage.

By the late 2010s the assumption had become implausible: coal growth had stalled, gas and renewables had displaced projected coal, and observed emissions were tracking well below the pathway. **Pielke and Ritchie documented the persistence of the business-as-usual label after the pathway had ceased to be a plausible baseline, and Hausfather and Peters made the same argument in *Nature* in 2020, titled to say it plainly.**

The correction has been substantially accepted within the field. It has not propagated to public communication at the same rate, and impact studies built on the pathway continue to be cited without the qualification.

This is the clearest case in the paper, because the correction came from inside and was published in the field's leading journal.

4. The Hot Model Problem

CMIP6, the model generation underlying AR6, contains models with equilibrium sensitivities above 5 degrees, higher than the previous generation and higher than paleoclimate evidence supports.

Hausfather, Marvel, Schmidt, Nielsen-Gammon and Zelinka addressed this in *Nature* in 2022, arguing that the hot models should not be used uncritically and that model results should be weighted by observational constraint rather than averaged. AR6 itself did not simply average CMIP6 output for its assessed warming, using constrained estimates instead.

The relevant point is not that the models are wrong. It is that the field identified a problem, published the correction in its most prominent venue, and changed its method, and **that this is a demonstration of the process working, which almost never appears in public communication because the communication has no register for it.**

5. Homogenisation of the Surface Record

Raw temperature observations require adjustment. Stations move, instruments change, observation times shift, and the time-of-observation bias in the United States record is real and documented. Adjusting for these is correct practice and refusing to adjust would produce a worse record.

Two facts about the adjustments are nonetheless worth stating because they are usually omitted.

The adjustments are documented, defensible in mechanism, and net upward in the United States record. Time-of-observation bias alone accounts for a substantial share, and the physical reason is specific: a large number of US stations shifted from afternoon to morning observation across the twentieth century, and afternoon readings double-count warm days. Correcting it raises the early record's apparent coolness relative to the late record. That is a real correction and refusing it would produce a worse dataset.

Sensitivity analysis should state the dataset, period, adjustment method and alternatives being compared. This paper has not established the absence of published sensitivity analyses and therefore does not use that absence as evidence.

The direction of an adjustment is not evidence of error by itself. Its mechanism and performance against independent information are the relevant tests. A criticism requires a specific competing reconstruction and quantified differences.

Berkeley Earth was founded partly to test whether these choices drove the result, was funded in part by people expecting they did, and concluded they did not. That is the strongest available answer to the objection and it should be cited by anyone raising it, including us.

6. The Tropical Troposphere

Models predict that warming in the tropical upper troposphere should exceed surface warming by a factor of roughly two, a consequence of moist adiabatic lapse rate adjustment. The amplification is a robust model result rather than a tuning artefact.

Satellite and radiosonde measurements have consistently shown less amplification than models predict. **Christy and colleagues have published this discrepancy repeatedly; Santer and colleagues have argued the gap is within combined observational and internal-variability uncertainty.** The dispute concerns error bars on difficult measurements and it remains live.

AR6 discusses the discrepancy. Public communication does not, and a reader who encounters it elsewhere encounters it as a suppressed finding rather than as an acknowledged open question, which is the worst possible framing for something the reports actually contain.

7. Extreme Event Attribution

AR6 assigns varying confidence to trends in extreme-weather categories, and the variation is large.

It reports high confidence in increased frequency and intensity of heat extremes. It reports considerably lower confidence for tropical cyclone frequency, for trends in meteorological drought in many regions, for river flooding, and for tornadoes. In several categories it states that observed trends are not detectable above natural variability, or that data are insufficient.

A low confidence assessment of a long term trend does not by itself invalidate attribution of a particular event. Event studies may evaluate a conditional change in probability or intensity. Compare the same variable, region, period and causal question before diagnosing an overstatement.

A proposed communication audit should preserve those distinctions and cite both the public statement and its scientific comparator. The assessment report alone cannot establish what public communicators said.

8. Why This Damages the Science

The overstatement is usually defended as necessary simplification for a public that will not absorb confidence intervals. The defence fails on its own terms.

Overstatement is discovered, and it is discovered by exactly the wrong audience. A reader disposed to doubt goes looking, finds that RCP8.5 was called business-as-usual after the field had stopped believing it, and does not conclude that a communicator simplified. He concludes he was handled, and he then discounts the entire body of work including the parts that are solid.

And the discovery is cheap. Every item in this paper is in the assessment reports or in *Nature*. None required specialist access. A motivated reader with an afternoon can find all six.

The asymmetry is what makes it costly. **A field's credibility is the one asset it cannot rebuild quickly**, and it is being spent to purchase rhetorical advantage on questions where the underlying evidence did not need the help.

9. What This Paper Does Not Argue

It does not argue that warming is not occurring. The instrumental record, the ocean heat content record, the sea level record and the cryosphere record are independent lines pointing the same way, and none of the six items above touches them.

It does not argue that CO is not a greenhouse gas. The absorption spectrum is laboratory physics from the nineteenth century.

It does not argue that the scientists involved are dishonest. **Every item documented here was identified and published by scientists inside the field**, which is the strongest available evidence that the process works and the clearest indictment of a communication practice that omits the corrections.

It does not argue that policy should change. That question involves discount rates, adaptation costs and value judgements outside the scope of anything argued here.

10. Recommendation

We propose one thing: that public communication carry the assessment reports' own confidence language rather than a compressed version.

This costs nothing. The language exists, it was constructed for this purpose, and using it requires no additional analysis. **A statement reading *the IPCC assigns low confidence to observed trends in this category* is not harder to write than one asserting the trend, and it has the advantage of being what the source says.**

The objection that audiences cannot handle uncertainty is untested and probably wrong. Audiences handle uncertainty in weather forecasts, medical prognosis and financial advice. What they handle badly is discovering that uncertainty was withheld.

Conclusion

The proposed audit compares public climate statements with the findings and uncertainty they cite. Each comparison needs a date, a matching scientific claim and the relevant conditions.

The assessment record includes progress as well as remaining uncertainty. Establishing a wider communication gap requires a defined sample and consistent coding, not a selection of striking examples. Publishing those comparisons would show where communication is accurate and where it needs correction.

References

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Hausfather, Z., & Peters, G. P. (2020). Emissions: the 'business as usual' story is misleading. *Nature*, 577, 618-620.

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Rohde, R., et al. (2013). Berkeley Earth temperature averaging process. *Geoinformatics & Geostatistics*, 1(2).

Verification status. The sensitivity range correction is checked against IPCC AR6 WGI SPM A.4.4, https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM.pdf. A representative communication sample and claim by claim source matrix are still required before reporting a systematic confidence gap. The absence of sensitivity analysis is not an established finding.

Approximately 2,000 words.